Task: Upgrade Windows Server 2016 VM to Windows Server 2022

Objective

To perform an in-place upgrade of an Azure Virtual Machine running Windows Server 2016 Datacenter to Windows Server 2022 Datacenter using Microsoft's Windows Server Upgrade process.

Requirements for the Task

- Azure Subscription
- Windows Server 2016 Virtual Machine
- Azure Cloud Shell (PowerShell)
- Administrative access to the VM
- Internet connectivity
- OS Disk Snapshot for rollback

Components Used

- Microsoft Azure Portal
- Windows Server 2016 Datacenter VM
- Azure Cloud Shell PowerShell
- OS Disk Snapshot
- Windows Server 2022 Upgrade Disk
- Windows PowerShell

Procedure

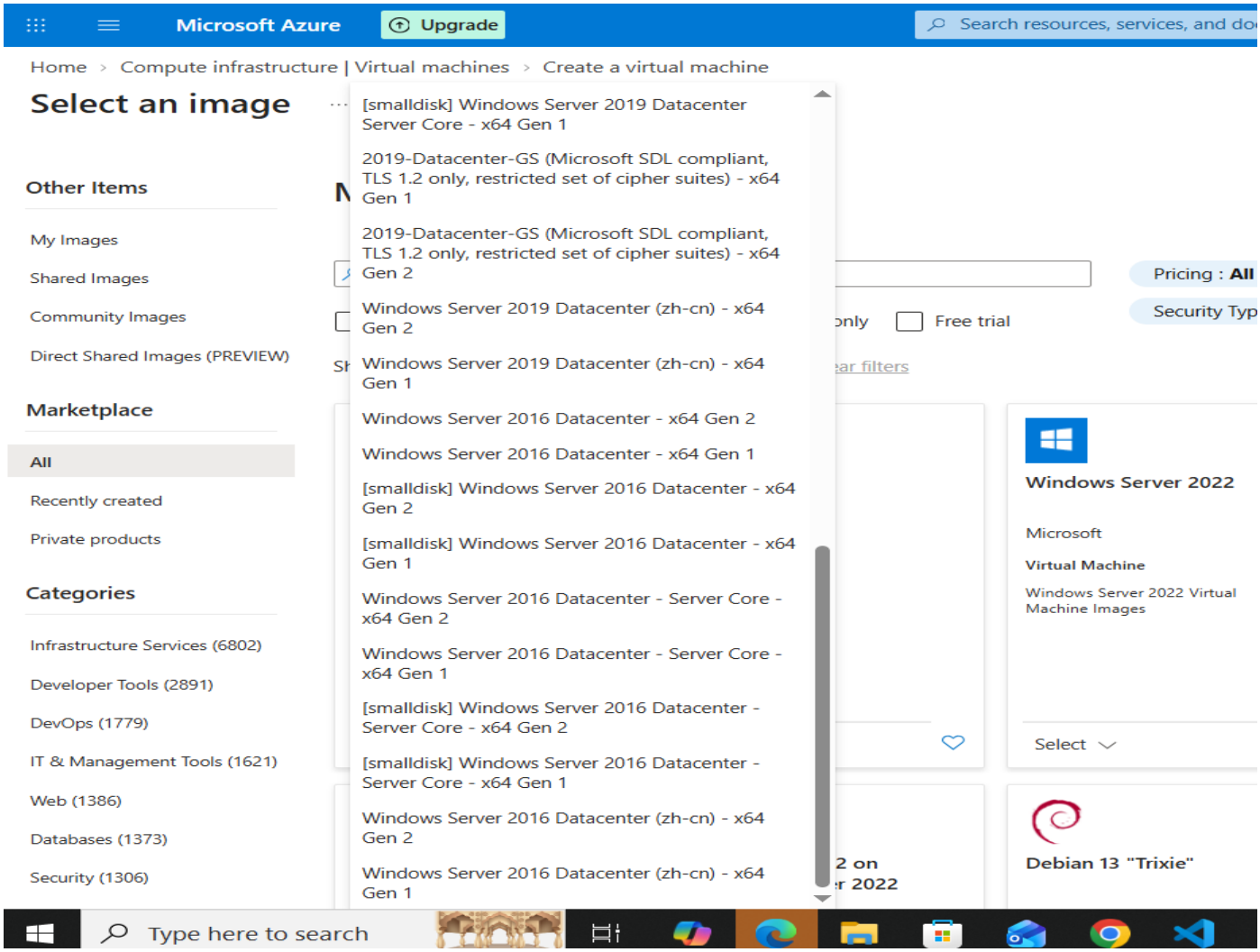
Step 1: Create a Windows Server 2016 Virtual Machine

1. Sign in to the Azure Portal.
2. Navigate to **Virtual Machines**.
3. Click **Create** → **Azure Virtual Machine**.

4. Select the image:

Windows Server 2016 Datacenter - x64 Gen 2

1. Configure VM settings and create the virtual machine. Windows Server 2016 image selection



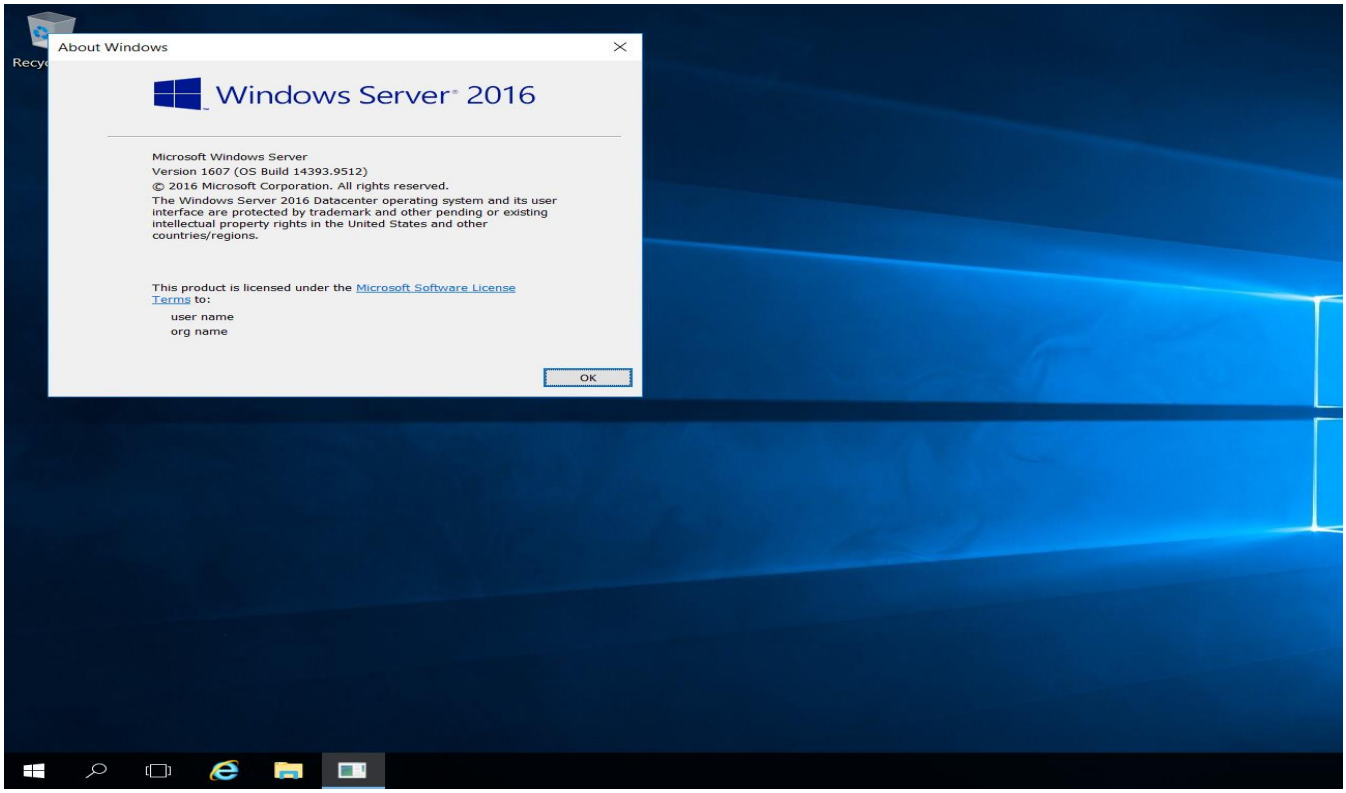
Step 2: Verify the Source Operating System

1. Connect to the VM using Remote Desktop Protocol (RDP).
2. Open Run and execute:

winver

1. Verify that the operating system is:

Windows Server 2016 Datacenter



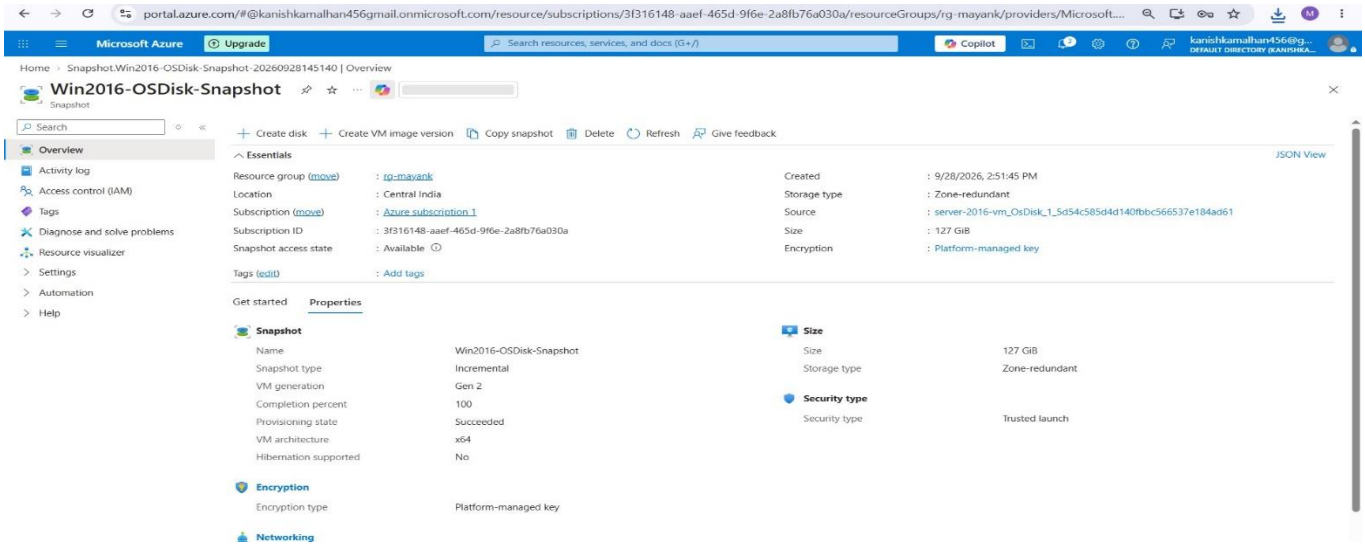
Step 3: Create an OS Disk Snapshot

1. Navigate to:

Virtual Machine → Disks

1. Select the OS Disk.
2. Click **Create Snapshot**.
3. Provide the snapshot name and create the snapshot.

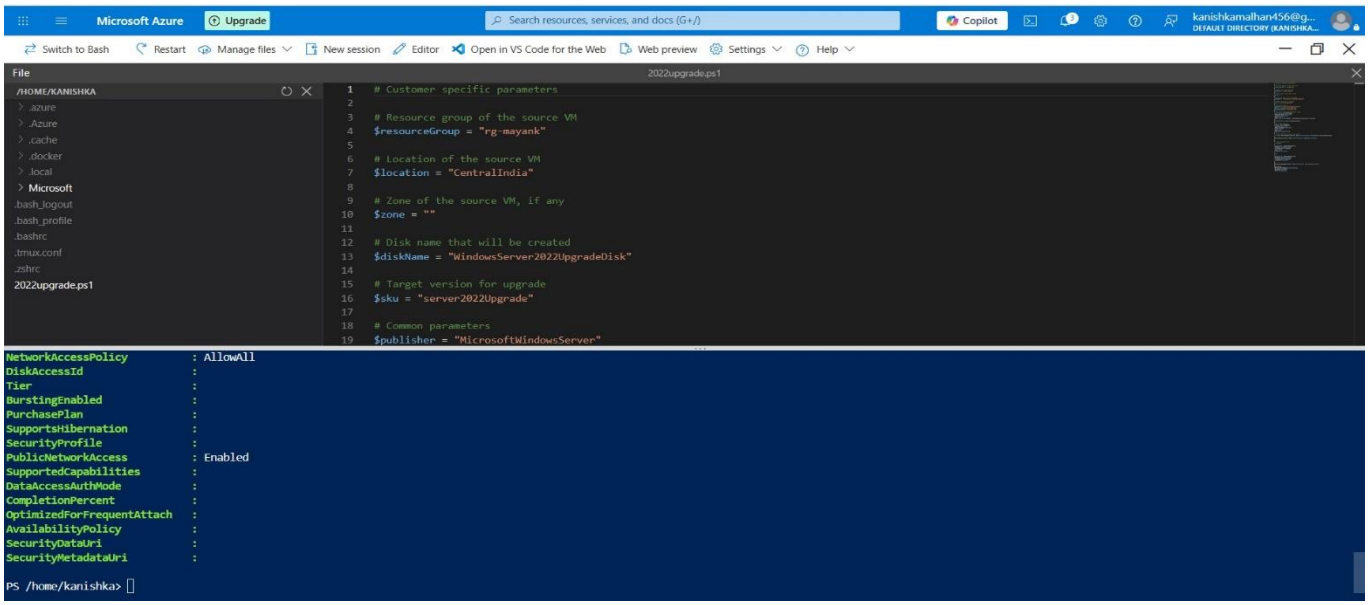
Purpose: The snapshot acts as a backup and allows rollback in case any issue occurs during the upgrade process.



Step 4: Configure the Windows Server 2022 Upgrade Script

1. Open **Azure Cloud Shell**.
2. Select **PowerShell**.
3. Open the PowerShell Editor.
4. Copy the Microsoft-provided Windows Server Upgrade script.
5. Modify the script according to the Azure environment:

resourceGroup = "rg-mayank"
location = "CentralIndia"
diskName = "WindowsServer2022UpgradeDisk"
sku = "server2022Upgrade"
Save the script.



Step 5: Execute the Upgrade Script

1. Run the PowerShell script.
2. Wait for successful execution.
3. Verify that a new managed disk is created.

Purpose: The script creates a Windows Server 2022 Upgrade Disk using Microsoft's hidden Azure Marketplace upgrade image.

Step 6: Attach the Upgrade Disk to the VM

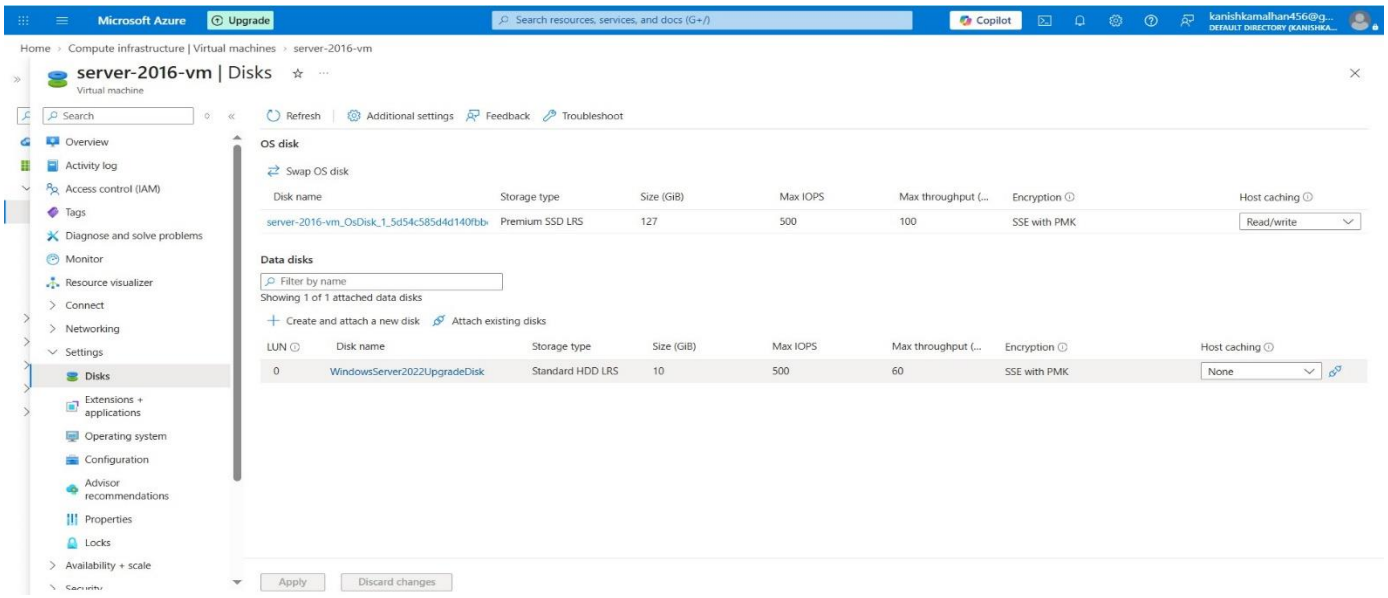
1. Open the virtual machine.
2. Navigate to:

Virtual Machine → Disks

1. Click **Attach Existing Disk**.
2. Select:

WindowsServer2022UpgradeDisk

1. Save the configuration.

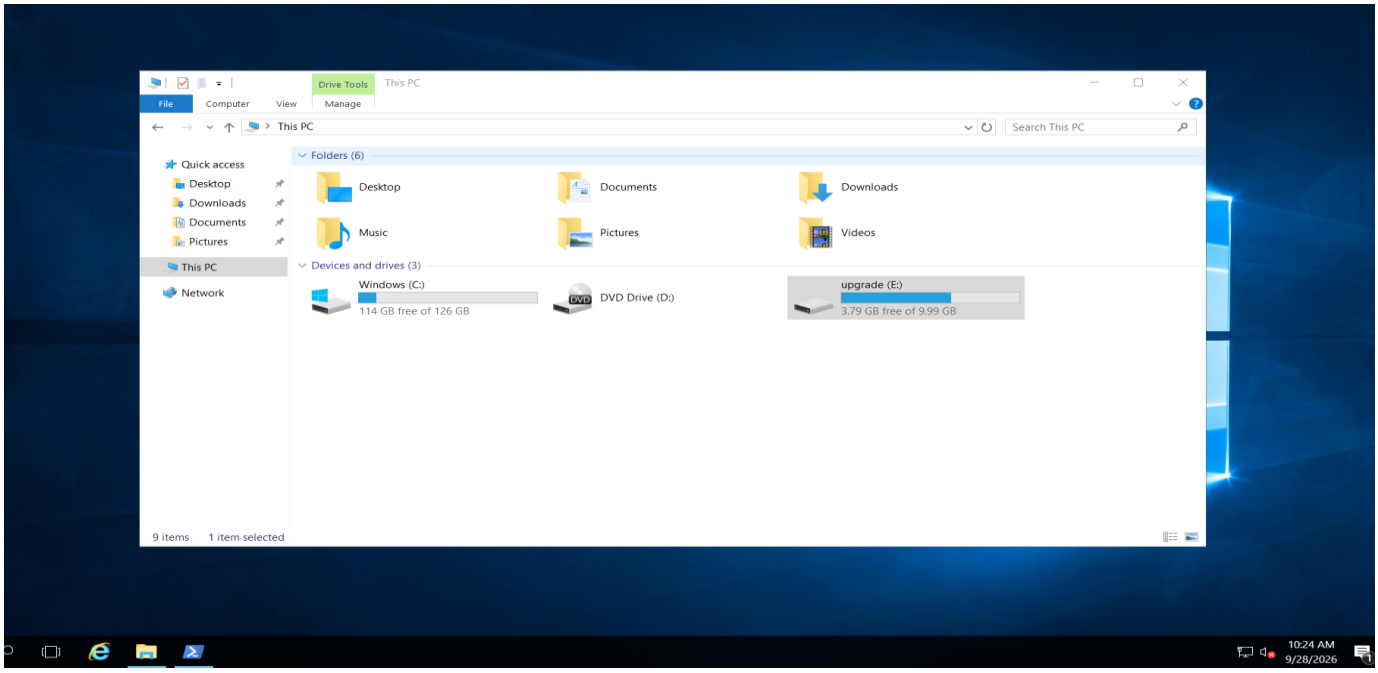


Step 7: Verify Upgrade Disk Availability

1. Connect to the VM through RDP.
2. Open **This PC**.
3. Confirm that the Upgrade Disk is visible.

Example:

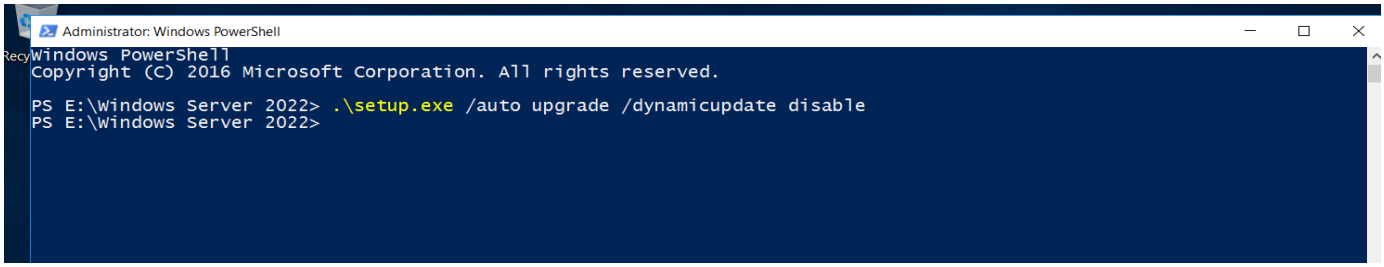
upgrade (E:)



Step 8: Start the In-Place Upgrade

1. Open **Windows PowerShell as Administrator**.
2. Navigate to the Windows Server 2022 installation directory.
3. Execute the following command:

.\setup.exe /auto upgrade /dynamicupdate disable
Purpose: Starts the automated in-place upgrade process.



Step 9: Configure Windows Setup

1. The Windows Setup wizard will start.
2. Select:

Windows Server 2022 Datacenter

1. Review the installation settings.
2. Proceed with the upgrade.

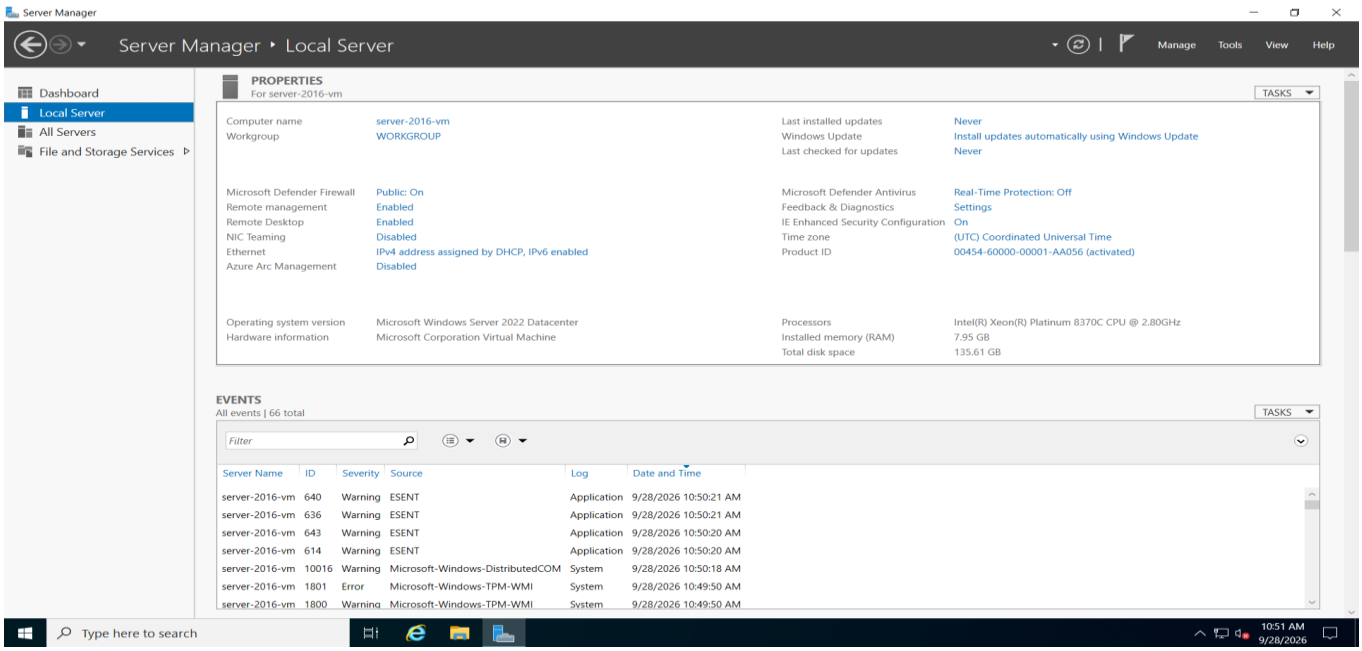
Step 10: Complete the Upgrade

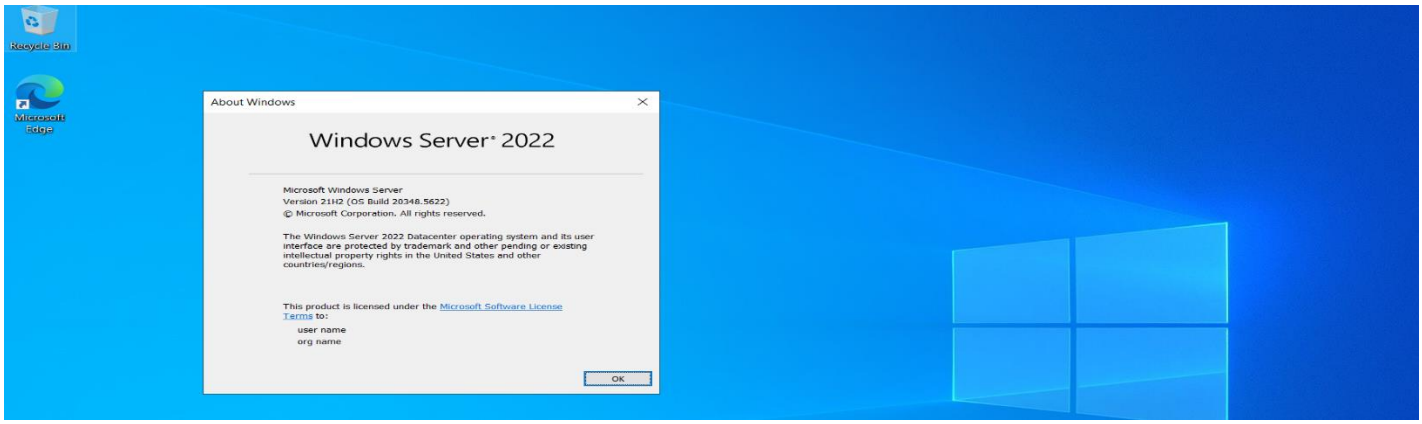
1. Allow compatibility checks to complete.
2. The installation process starts automatically.
3. The VM restarts multiple times during the upgrade process.
4. Wait until installation finishes successfully.

Step 11: Verify Upgrade Completion

1. Log in to the upgraded VM.
2. Open **Server Manager → Local Server**.
3. Verify that the operating system displays:

Microsoft Windows Server 2022 Datacenter





Observation

- Windows Server 2016 Datacenter VM was created successfully.
  - Source operating system was verified using the winver command.
  - OS Disk Snapshot was created successfully before performing the upgrade.
  - Microsoft Upgrade Script was configured and executed through Azure Cloud Shell.
  - Windows Server 2022 Upgrade Disk was successfully created.
  - Upgrade Disk was attached to the virtual machine.
  - Upgrade media became available inside the VM as an additional drive.
  - In-place upgrade was initiated through PowerShell.
  - Windows Server 2022 Datacenter edition was selected during setup.
  - The upgrade process completed successfully after multiple automatic restarts.
  - Existing VM configuration and data were retained during the upgrade.

Result

Successfully upgraded the Azure Virtual Machine from **Windows Server 2016 Datacenter** to **Windows Server 2022 Datacenter** using Microsoft's in-place upgrade method. The upgrade was completed successfully while preserving the existing virtual machine, configuration, and operating environment.